Robert Ryan

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Software engineer and IB Diploma student. This summer I built the software that kids with concussions use to see their doctor and test their eyes from home, at a pediatric telehealth startup in San Francisco. Quant ML research accepted at **ACM ICAIF '26**.

EXPERIENCE

Software Engineer · Headquarters Health, San Francisco — Jun 2026 – Present

- Merged **273 pull requests** (+319k lines, 77k of them tests) across nine production systems on a HIPAA-compliant pediatric concussion-care platform used by clinicians, patients and **10,000+ parents** (Next.js, React, AWS).
- Replaced Google Meet with an in-house telehealth platform on the Amazon Chime SDK; put the chart note and live-transcript AI drafting inside the visit, with per-participant transcription consent.
- Architected a camera-based concussion baseline: webcam saccade and smooth-pursuit tests, MediaPipe gaze analysis on AWS Lambda, presigned-S3 clip delivery, and a clinician view scoring patients against pooled norms. Webcam prototype to production in under a week.
- Grew one autonomous AI employee into a fleet of four on a harness built around headless Claude Code, with multi-model failover, a cross-model advisory panel and tiered memory; the agents authored **101 PRs** (87 merged) that I reviewed.

Computer Vision Research Assistant · NC State University, Proactive Pig Finishing Project — Jan – May 2026

- Designed a dual-branch CNN (pretrained ResNet18 fused with custom thermal-input layers) for real-time stress detection in a working livestock facility: **97% validation accuracy** on noisy, unconstrained thermal imagery. Built the preprocessing and synthetic-augmentation pipeline behind its robustness to occlusion, distance and sensor noise.

AI Systems Engineer · entirefeed.com — Mar – May 2026

- Built low-latency generative video pipelines, tuning inference throughput and GPU utilization for high-volume production, plus neural speech synthesis for voice cloning. Trained a vision-transformer model to predict TikTok views.

RESEARCH

Conformal Kelly · arXiv:2608.01494 · sole author, accepted at ACM ICAIF '26 (main conference); presenting a poster in Milan, Nov 2026 — Aug 2026

- Conformal prediction intervals as the scale in fractional-Kelly position sizing: split-conformal intervals around ridge-regression forecasts of 21-day ETF returns, in backtests with trading costs, execution lag and leverage limits.

Concussions in Plain Sight: Impaired Play and Scoreboard-Dependent Removal After Head Hits in the NFL · sole author, abstract accepted at MIT Sloan Sports Analytics Conference 2027 from a record number of submissions; invited to submit the full paper — Oct 2026

- Pre-registered event study over 17 NFL seasons (sealed 2024–25 holdout) detecting unreported concussions in public play-by-play data: 23% of concussed QBs finished the game, falling -0.33 EPA/dropback after the hit vs hit-but-undiagnosed controls ($p < 0.001$).
- Removals came more often once the game was already decided. Done during my internship at Headquarters Health, a pediatric concussion-care startup.

PROJECTS

- **Informed** (founder): iOS app that fact-checks short-form video; a few hundred users. informed-app.com
- **TurboEdit** (co-founder): web-based agentic video editor. turboedit.ai
- **nlsh**: natural-language shell in Rust, running fully on-device via Apple Foundation Models.
- **Neural Net Visualizer**: interactive 3D visualization of neural network architectures.

EDUCATION

ACS Athens, Greece · IB Diploma, class of 2027 · Predicted 43/45 · GPA 4.0 unweighted · SAT 1540 · Dual-enrollment college GPA 4.0

ACHIEVEMENTS

1st place ($3,000), Zero Human Hackathon, Athens (built an AI property manager) · 2nd place, Cursor Hackathon (built a multiplayer typing-race app) · Finalist, Qualcomm × Meta ExecuTorch Hackathon · Selected, Y Combinator Startup School 2026 · Coding Award, aerial drone competition · Oklahoma State Science Fair winner · TED-Ed speaker

SKILLS

TypeScript · React / Next.js · Python · SQL · Rust · AWS (Lambda, S3, Chime SDK, Amplify) · MediaPipe · CNNs · TensorFlow Lite · PX4 / ArduPilot · Three.js / GLSL